

Project Plan: Whisper Speech-to-Text Module

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Objective

To develop and integrate a speech recognition module using OpenAI's Whisper API that accepts audio input (.wav/.mp3), transcribes it to text, and sends the result to the backend via REST API or WebSocket.

Task Breakdown and Descriptions

Task	Description
Development	Implement core functionality: audio input, transcription using Whisper API, multi-turn dialogue support, and data return via REST/WebSocket.
Testing & Debugging	Unit testing with various audio samples, simulate noisy inputs, and correct functional errors.
Team Evaluation	Share the module with team for code review, usability testing, and feedback.
Final Overall Testing	Conduct integrated testing with backend, evaluate full transcription and response loop.
Documentation & Cleanup	Write developer guide, API usage summary, and clean up the codebase for maintainability.
Buffer Time	Reserve time for unforeseen issues, bug fixes, or last-minute improvements.

Time Estimation (in Days)

Task	Duration (Days)
Development	4 days
Testing & Debugging	2 days
Team Evaluation	2 days
Final Overall Testing	1 days
Documentation & Cleanup	1 day
Buffer Time	1 day
Total Estimated Time	11 days

Task Dependencies

- Development must be completed before Testing & Debugging.- Team Evaluation follows after the module is tested.- Final Testing can only start post team evaluation.- Documentation comes after all technical validation is done.- Buffer time is used at the very end.

Schedule (Starting from Day 1)

Day	Task	Notes
1	Development	Setup, API integration, audio input
2	Development	Format handling, Whisper logic
3	Development	Multi-turn support, response output
4	Development	REST/WebSocket integration, cleanup
5	Testing & Debugging	Unit tests, short/long clips
6	Testing & Debugging	Simulate errors, edge case handling
7	Team Evaluation	Code review, feedback
8	Team Evaluation	Fixes based on review
9	Final Overall Testing	Backend integration tests
10	Final Overall Testing	Full flow verification
11	Documentation & Cleanup	Write docs, finalize API details
12	Buffer Time	Fixes, polish, backup

Gantt Chart Overview

